

Headline verdict — 2026-05-05 re-run

Mostly pass · 3 triangulated discrepancies · K⁺ alternative still open · cross-tool coverage strengthened on T03

1SAR is a real and substantial improvement over the deposited starting model.
Modelling decisions on structure, geometry, ion identity (Mg²⁺ excluded), water, disulfides — all defensible.

Closed across the 04-30 / 05-01 / 05-04 / 05-05 re-issues

- ✓ Mg²⁺ alternative ruled out — CheckMyMetal-style IZI 16.35
- ✓ Ligand / metal density support confirmed — RSCC ≥ 0.96
- ✓ T13 data-quality cross-tool gap closed — ctruncate (05-04)
- ✓ T03 strengthened — Servalcat sigmaa direct R-work/R-free, 4-tool bracket (NEW 05-05)

Still discrepant (3-tool confirmed)

- x R-free reproducibly ≈ 0.21 (4-tool bracket), agent reports 0.199
- x Round 6 R-free 0.207 < round 7 R-free 0.212
- x Asn A 39 is a Ramachandran outlier
- x Density-peak inventory incomplete (23+ / 8– vs 9 listed)

Open question

K⁺ alternative for both ions
K⁺ has the best raw bond-length fit (IZI 0.13–0.44) but coordination number is too low for K⁺ canonical, and the protein is a known Ca²⁺ enzyme. Needs crystallisation buffer information.

Net acceptance against agent’s own success criteria — 3 / 6 pass cleanly

R-free improvement ✓ pass	R-free gap < 0.05 ⚠ 3 of 4 oracles fail	Ramachandran < 1% ✓ headline wrong	Clashscore < 5 ✓ pass	No Δρ > 4σ x fails	Data quality (T13) ✓ pass
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